1'0
BUF
ap_ST_fsm_state4_blk
ap_ST_fsm_state3_blk
ap_ST_fsm_state1_blk
J_edges_0
$161436
$mux
CLK
D
Q
$163181
$dff
J_edges_0_read_reg_587
J_edges_27
$161382
$163208
J_edges_27_read_reg_722
J_edges_26
$161384
$163207
J_edges_26_read_reg_717
J_edges_17
$161402
$163198
J_edges_17_read_reg_672
J_edges_20
$161396
$163201
J_edges_20_read_reg_687
J_edges_16
$161404
$163197
J_edges_16_read_reg_667
J_edges_25
$161386
$163206
J_edges_25_read_reg_712
$163171
$161380
e_new_reg_727
sext_ln257_fu_469_p1
J_edges_19
$161398
$163200
J_edges_19_read_reg_682
J_edges_24
$161388
$163205
J_edges_24_read_reg_707
J_edges_23
$161390
$163204
J_edges_23_read_reg_702
J_edges_18
$161400
$163199
J_edges_18_read_reg_677
J_edges_15
$161406
$163196
J_edges_15_read_reg_662
J_edges_1
$161434
$163182
J_edges_1_read_reg_592
J_edges_22
$161392
$163203
J_edges_22_read_reg_697
ap_CS_fsm_state1
ap_CS_fsm_state2
ap_CS_fsm_state3
ap_CS_fsm_state4
ap_CS_fsm_state5
J_edges_14
$161408
$163195
J_edges_14_read_reg_657
J_edges_13
$161410
$163194
J_edges_13_read_reg_652
5'00010
5'00100
$161361
10'0100010000
4:0 - 24:20
4:0 - 19:15
9:0 - 14:5
4:0 - 4:0
grp_dense_energy_over_alpha_fu_379_ap_done
$24
$and
5'xxxxx
$161366
$pmux
ap_NS_fsm
5'00001
$161473
$163211
ap_CS_fsm
$161371_CMP0
$161370_CMP0
$161367_CMP0
$161369_CMP0
$161368_CMP0
$eq
5'10000
5'01000
0:0 - 4:4
0:0 - 3:3
0:0 - 2:2
0:0 - 1:1
0:0 - 0:0
grp_dense_energy_over_alpha_fu_379_ap_return
new_energy
ap_rst
ap_clk
J_edges_0_val
J_edges_10_val
J_edges_11_val
J_edges_12_val
J_edges_13_val
J_edges_14_val
J_edges_15_val
J_edges_16_val
J_edges_17_val
J_edges_18_val
J_edges_19_val
J_edges_1_val
J_edges_20_val
J_edges_21_val
J_edges_22_val
J_edges_23_val
J_edges_24_val
J_edges_25_val
J_edges_26_val
J_edges_27_val
J_edges_2_val
J_edges_3_val
J_edges_4_val
J_edges_5_val
J_edges_6_val
J_edges_7_val
J_edges_8_val
J_edges_9_val
ap_clk
ap_rst
ap_start
h_0_val
h_1_val
h_2_val
h_3_val
h_4_val
h_5_val
h_6_val
h_7_val
spins
grp_dense_energy_over_alpha_fu_379
uniform_move_operation_dense_energy_over_alpha
ap_done
ap_idle
ap_ready
ap_return
grp_dense_energy_over_alpha_fu_379_ap_idle
grp_dense_energy_over_alpha_fu_379_ap_ready
J_edges_12
$161412
$163193
J_edges_12_read_reg_647
J_edges_11
$161414
$163192
J_edges_11_read_reg_642
J_edges_10
$161416
$163191
J_edges_10_read_reg_637
J_edges_21
$161394
$163202
J_edges_21_read_reg_692
J_edges_2
$161432
$163183
J_edges_2_read_reg_597
1:1 - 0:0
h_7
$161438
$163180
h_7_read_reg_582
old_energy
8:8 - 9:9
8:0 - 8:0
sext_ln257_1_fu_472_p1
$161456
h_6
$161440
$163179
h_6_read_reg_577
h_5
$161442
$163178
h_5_read_reg_572
h_4
$161444
$163177
h_4_read_reg_567
1'1
$161459
new_spins
$161454
$163172
new_spins_read_reg_542
J_edges_7
$161422
$163188
J_edges_7_read_reg_622
grp_dense_energy_over_alpha_fu_379_ap_start_reg
grp_dense_energy_over_alpha_fu_379_ap_start
h_3
$161446
$163176
h_3_read_reg_562
h_2
$161448
$163175
h_2_read_reg_557
h_1
$161450
$163174
h_1_read_reg_552
h_0
$161452
$163173
h_0_read_reg_547
$163209
$161462
J_edges_6
$161424
$163187
J_edges_6_read_reg_617
J_edges_9
$161418
$163190
J_edges_9_read_reg_632
J_edges_8
$161420
$163189
J_edges_8_read_reg_627
J_edges_5
$161426
$163186
J_edges_5_read_reg_612
J_edges_4
$161428
$163185
J_edges_4_read_reg_607
J_edges_3
$161430
$163184
J_edges_3_read_reg_602
3:3 - 0:0
2:2 - 0:0
4:4 - 0:0
$31
$sub
sub_ln257_fu_476_p2
9:9 - 0:0
8:8 - 0:0
$34
$xor
xor_ln257_1_fu_514_p2
$35
xor_ln257_fu_502_p2
$28
$and
and_ln257_fu_508_p2
9'100000000
9'011111111
$30
select_ln257_fu_520_p3
$33
x_positive_part_fu_458_delta_over_alpha
tmp_fu_482_p3
delta_over_alpha_fu_490_p1
tmp_1_fu_494_p3
ap_rst
delta_over_alpha
x_positive_part_fu_458
uniform_move_operation_positive_part
x_positive_part_fu_458_ap_ready
x_positive_part_fu_458_ap_return
0 -> 8:8
7:0 - 7:0
delta_x
$161352
$27
grp_chebyshev_g_fu_463_ap_done
ap_clk
ap_start
x
grp_chebyshev_g_fu_463
uniform_move_operation_chebyshev_g
grp_chebyshev_g_fu_463_ap_idle
grp_chebyshev_g_fu_463_ap_ready
grp_chebyshev_g_fu_463_ap_return
$161464
accept_prob
grp_chebyshev_g_fu_463_ap_start_reg
grp_chebyshev_g_fu_463_ap_start
$161378
$163170
x_reg_732
$161467
$161470
$163210

uniform_move_operation